

Quantization Beam Analysis and Codebook Design for One-Bit Reconfigurable Intelligent Surface

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Abstract—The 1-bit phase controlling of the reconfigurable intelligent surface (RIS) is commonly leveraged to reduce deployment costs. However, the 1-bit phase limitation also brings an unavoidable quantization beam on passive beamforming. In this letter, we investigate the quantization beam on a conventional 1-bit RIS and design the prephasing map to suppress the power of the quantization beam while enhancing passive beamforming. In particular, we analytically derive directions of the quantization beam of 1-bit RIS, which is important for interference management and multi-user broadcasting. Moreover, we propose the map fusion (MF) method to improve the power gain in the RIS codebook design while suppressing the power of the quantization beam. The simulation result shows that directions of the quantization beam can be precisely obtained, and the proposed MF method can effectively suppress the negative impact of the quantization beam and improve transmission efficiency.

Index Terms—Reconfigurable intelligent surface, beamforming, mmWave, quantization beam, 1-bit.

I. INTRODUCTION

IN RECENT years, utilizing reconfigurable intelligent surface (RIS) has emerged as a strategic approach to improve power and spectrum utilization. This concept hinges on the fundamental principle of efficiently collecting transmitted energy of electromagnetic (EM) wave and precisely guiding it to the desired location [1]. While the roadmap for RIS holds great promise, the RIS implementation is encountering challenges due to practical constraints.

The practical constraint of RIS mainly comes from the specific hardware implementation of individual elements [2]. Due to limited cost, applying 1-bit RIS is the most feasible choice as a deployment plan to a large scale. Thus, relative works have proposed solutions to beamforming and channel estimation while considering 1-bit phase controlling. For instance, [2] proposed a low-complexity channel estimation scheme for orthogonal frequency-division multiplexing (OFDM) transmission system with 1-bit RIS. This letter [3] jointly designed the transmit beamformer and RIS reflection to maximize the

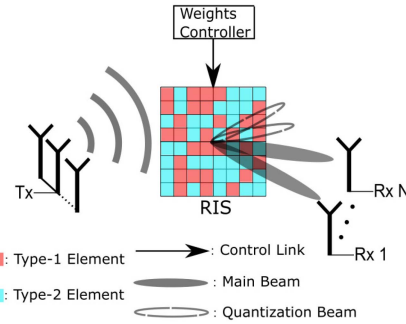


Fig. 1. 1-bit RIS with a typical prephasing map.

average sum rate. However, [2] and [3] overlooked the physical reflection characteristic on 1-bit RIS, which is the quantization beam. Specifically, the quantization beam is the additional beam with equal power to the main beam in 1-bit RIS. The works [4], [5], [6] have physically confirmed the quantization beam phenomenon of 1-bit RIS, with only one type of binary phase-switching elements by hardware experiments. Note that the quantization beam causes additional interference to the wireless environment and ambiguity during sensing [7] and channel estimation [8]. In this regard, [6] proposed random prephasing to mitigate the quantization beam's power in order to maintain a single-beam pattern. In particular, the random prephasing is equivalently achieved by randomly selecting a combination of two types of elements¹ on RIS, as shown in Fig. 1. However, the analysis in [6] only indicates that the reason for the quantization beam is the symmetry of beam pattern with normally incident plane waves. To the best of the authors' knowledge, the analytical direction of the quantization beam with different directions of the incident wave is still unknown. In addition, random prephasing design also introduced uncontrollable beamforming gain for all steered directions. This letter's main contributions are summarized as follows.

- The in-depth cause of the quantization beam is analyzed, and the direction of the quantization beam is derived in closed-form expressions. Simulations have verified the accuracy of the proposed corollaries.
- A codebook design problem is formulated to maximize the sum capacity at multiple directions, which is solved by the proposed map fusion (MF) method. Simulation results show an improvement in beamforming power gain over the current random prephasing method.

¹These two types of 1-bit elements have a different phase-switching statement with the largest Euclidean distance on the complex plane. E.g., type-1 elements can switch between 0° and 180° on a constant modulus circle. In comparison, type-2 elements can switch between 90° and 270° .

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II. SYSTEM MODEL

A. The Single RIS Channel Model

The considered system contains one transmitter (Tx) with N_{Tx} transmit antennas and a total of N_{Rx} receivers, where each receiver (Rx) equips a single antenna at different directions. A single RIS is assumed to be the only reflector in the transmission environment and is composed of total M elements, as shown in Fig. 1. We consider line-of-sight (LoS) channels obeying the quasi-optical transmission nature to be consistent with the practical situation in millimeter wave (mmWave). Incident directions from antennas on Tx towards RIS are considered of the same direction $\Omega_{in,Tx}$ while the exit directions of Tx phased array towards RIS are considered of the same direction $\Omega_{out,RIS}$. Thus, the channel between RIS and Tx is a rank-one channel [9] and can be represented by

$$\mathbf{E} = \sqrt{l_p} \mathbf{a}(\Omega_{in,Tx}) \mathbf{a}(\Omega_{out,RIS})^H, \quad (1)$$

where $\mathbf{a}(\Omega)$ is the steering vector of direction Ω ². Note that Ω_{in}/Ω_{out} means the spatial information of incident/exit direction of RIS. Moreover, l_p is the path loss factor for LoS path whose specific expression can be referred to in [10]. Denote $\mathbf{A}_{out} \in \mathbb{C}^{M \times N_{Rx}}$ as the LoS channel matrix of angle of departure (AOD) from RIS to receivers. Then, there is

$$\mathbf{A}_{out} = \sqrt{l_p} [\mathbf{a}(\Omega_{out,1}), \mathbf{a}(\Omega_{out,2}), \dots, \mathbf{a}(\Omega_{out,N_{Rx}})], \quad (2)$$

where $\Omega_{out,i}$ is exit direction from RIS to Rx_{*i*}. Consequently, the received signal for receivers can be written as

$$\mathbf{y} = \mathbf{A}_{out}^T \mathbf{W} \mathbf{E} \mathbf{P}_o \mathbf{s} + \mathbf{n}, \quad (3)$$

where $\mathbf{s} = [s_1, s_2, \dots, s_{N_{Tx}}]^T \in \mathbb{C}^{N_{Tx} \times 1}$ is the source signal vector and $\mathbf{P}_o \in \mathbb{C}^{N_{Tx} \times N_{Tx}}$ is the precoding matrix at Tx. In addition, $\mathbf{n} \in \mathbb{C}^{N_{Rx} \times 1}$ is the noise vector at receivers, and the RIS weights matrix $\mathbf{W} \in \mathbb{C}^{M \times M}$ is a diagonal matrix with each entity on the diagonal being the weight value. We denote $\mathbf{w}$ as a column vector whose elements are the main diagonal elements of $\mathbf{W}$ and denote the quantized weights vector as $\mathbf{w}_q \in \mathbb{C}^{M \times 1}$. For 1-bit RIS with two types' 1-bit elements, we have

$$w_{q,m} = \begin{cases} e^{j0} \text{ or } e^{j\pi}, & z_m = 0 \\ e^{-j\frac{\pi}{2}} \text{ or } e^{j\frac{\pi}{2}}, & z_m = 1, \end{cases} \quad (4)$$

where the value of z_m is binary and decides whether m -th element belongs to type-1 or type-2. The prephasing map vector, $\mathbf{z} \in \mathbb{C}^{M \times 1}$, needs to be designed prior to the production of RIS and can not be changed after the production.

B. Transmission Model

To avoid multi-user (MU) interference [11], transmission has to consume extra frequency bands or time blocks. With

²The spatial information Ω contains azimuth and elevation angles for two-dimensional deployment (e.g., uniform rectangular array (URA)), or contains only one-dimensional information of azimuth angle such as uniform linear array (ULA).

division on time/frequency, each transmission symbol is individual between receivers, and the received signal of Rx_{*i*} in Eq. (3) can be rewritten as

$$y_i = \mathbf{a}_{\Omega_{out,i}}^T \mathbf{W} \mathbf{E} \mathbf{p}_i s_i + n_i, i = 1, 2, \dots, N_{Rx}. \quad (5)$$

Moreover, by applying maximum ratio combining (MRC) beamformer at Tx, Eq. (5) can be simplified to

$$y_i = \sqrt{P_{Tx}} \mathbf{w}^H \mathbf{a}_C(\Omega_{out,i}, \Omega_{in,Tx}) s_i + n_i, \quad (6)$$

where P_{Tx} is the power gain from MRC beamforming at Tx and

$$\mathbf{a}_C(\Omega_{out,i}, \Omega_{in,Tx}) = \mathbf{a}(\Omega_{out,i}) \odot \mathbf{a}(\Omega_{in,Tx}), i = 1, \dots, N_{Rx}, \quad (7)$$

which is the combined steering vector from the incident and exit steering vector of RIS.

III. QUANTIZATION BEAM ANALYSIS

For narrative simplicity, we consider a ULA-type RIS with element spacing $d < \frac{\lambda}{2}$ and unified path loss. Thus, the combined steering vector has a form of

$$\mathbf{a}_C(\phi_{out,i}, \phi_{in,Tx}) = \begin{bmatrix} 1, e^{-jkd(\cos\phi_{in,Tx} + \cos\phi_{out,i})}, \\ \dots, e^{-jkd(\cos\phi_{in,Tx} + \cos\phi_{out,i})(M-1)} \end{bmatrix}^T, \quad (8)$$

where $k = \frac{2\pi}{\lambda}$ is the wavenumber and ϕ is the azimuth angle of linear array. Given only one type of 1-bit elements, $\mathbf{z} = \mathbf{1}$ or $\mathbf{0}$, the quantization beam has equal power with the main beam. Then, we have

$$\left| \mathbf{w}_q^H \mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) \right| = \left| \mathbf{w}_q^H \mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}) \right|, \quad (9)$$

where the direction of the main beam is denoted as $\phi_{out,MB}$ while the direction of quantization beam is denoted as $\phi_{out,Q}$ and $\phi_{out,MB} \neq \phi_{out,Q}$. Then, we infer the reason for the quantization beam. Disregarding the design of $\mathbf{w}$, the following condition is critical to satisfy Eq. (9) if

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = \begin{cases} \pm \mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}) \\ \pm \mathbf{a}_C^*(\phi_{out,Q}, \phi_{in,Tx}), \end{cases} \quad (10)$$

where $*$ is the conjugate symbol.

Lemma 1: Within the vector space of Eq. (9), there is

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) \neq -\mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}) \quad \text{and} \quad -\mathbf{a}_C^*(\phi_{out,Q}, \phi_{in,Tx}). \quad (11)$$

Proof: Assume

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = -\mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}), \quad (12)$$

and then there is

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = e^{j\pi} \mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}). \quad (13)$$

By taking the m -th term in Eq. (13), we obtain

$$\begin{aligned} & kd(\cos\phi_{out,MB} + \cos\phi_{in,Tx})m \\ &= kd(\cos\phi_{out,Q} + \cos\phi_{in,Tx})m + \pi + 2\pi n, \end{aligned} \quad (14)$$

where $n \in \mathbb{Z}$. For $m = 0$, Eq. (14) can not be satisfied, which indicates that the original assumption does not hold. Similarly, the assumption of m -th term equality regarding $-\mathbf{a}_C^*(\phi_{out,Q}, \phi_{in,Tx})$ can not hold and the proof is completed. ■

Lemma 2: Within the vector space of Eq. (8), there exists

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = \mathbf{a}_C^*(\phi_{out,Q}, \phi_{in,Tx}). \quad (15)$$

Proof: By checking the m -th term in Eq. (15), there is

$$\begin{aligned} kd(\cos \phi_{out,MB} + \cos \phi_{in,Tx})m \\ = -kd(\cos \phi_{out,Q} + \cos \phi_{in,Tx})m - 2\pi n. \end{aligned} \quad (16)$$

Then, by extracting $\phi_{out,Q}$, there is

$$\phi_{out,Q} = \arccos\left(-\cos \phi_{out,MB} - 2\cos \phi_{in,Tx} - \frac{2\pi n}{kdm}\right). \quad (17)$$

As long as the value of $(-\cos \phi_{out,MB} - 2\cos \phi_{in,Tx} - \frac{2\pi n}{kdm})$ locates within the feasible domain $[-1, 1]$, there is $\phi_{out,Q}$ and the conjugate symmetry of Eq. (15) exist. ■

Lemma 3: Given $d < \frac{\lambda}{2}$, there is

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) \neq \mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx}). \quad (18)$$

Proof: Assume equality of Eq. (18) holds, and therefore for equality of m -th term, there is

$$\begin{aligned} kd(\cos \phi_{out,MB} + \cos \phi_{in,Tx})m \\ = kd(\cos \phi_{out,Q} + \cos \phi_{in,Tx})m + 2\pi n. \end{aligned} \quad (19)$$

As the first two terms in Eq. (19) are equal when $m = 0$, consider $m = 1, 2, \dots, M-1$, there are

$$\cos \phi_{out,MB} = \cos \phi_{out,Q} + \frac{2\pi n}{kdm} \quad (20)$$

and

$$\phi_{out,MB} = \arccos\left(\cos \phi_{out,Q} + \frac{2\pi n}{kdm}\right). \quad (21)$$

Note when $n = 0$, Eq. (21) is contradicted with $\phi_{out,MB} \neq \phi_{out,Q}$. Thus, for $n \neq 0$ and $d < \frac{\lambda}{2}$, there is

$$\frac{2\pi n}{kdm} \in (-\inf, -2) \cup (2, \inf). \quad (22)$$

Therefore, we obtain

$$\left(\cos \phi_{out,Q} + \frac{2\pi n}{kdm}\right) \in (-\inf, -1) \cup (1, \inf), \quad (23)$$

which shows the original assumption does not hold. The proof is completed. ■

By leveraging Lemma 1, Lemma 2, and Lemma 3, we know the conjugate equality

$$\mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = \mathbf{a}_C^*(\phi_{out,Q}, \phi_{in,Tx}) \quad (24)$$

exist within element spacing less than half wavelength³. Moreover, considering $\mathbf{w}_q$ with map vector $\mathbf{z} = \mathbf{0}$, we know

$$\mathbf{w}_q^H \mathbf{a}_C(\phi_{out,MB}, \phi_{in,Tx}) = \left(\mathbf{w}_q^H \mathbf{a}_C(\phi_{out,Q}, \phi_{in,Tx})\right)^* \quad (25)$$

³For element spacings greater than half wavelength, the in-phase equality in Lemma 3 exists but introduces grating lobes.

can always be satisfied, which is a sufficient condition to satisfy Eq. (9). Therefore, Eq. (25) means the combined steering vector and quantized weights jointly produce the quantization beam⁴. With the conjugate equality-induced direction of quantization beam in Eq. (17), there are the following corollaries by classification:

Corollary 1: When $d \in (0, \lambda/4)$, the quantization beam's direction ϕ_Q can be calculated by

$$\phi_Q = \begin{cases} \arccos(\gamma_Q), & \gamma_Q \in [-1, 1] \\ \emptyset, & \text{else,} \end{cases} \quad (26)$$

where $\gamma_Q = -2\cos(\phi_{in,Tx}) - \cos(\phi_{out,MB})$.

Corollary 2: When $d \in [\lambda/4, \lambda/2)$, the quantization beam's direction ϕ_Q can be calculated by

$$\phi_Q = \begin{cases} \arccos(\gamma_Q), & \gamma_Q \in [-1, 1] \\ \arccos(\gamma_Q + \lambda/d), & \gamma_Q \in (-3, -1), \lambda/d \leq 1 - \gamma_Q \\ \emptyset, & \gamma_Q \in (-3, -1), \text{else} \\ \arccos(\gamma_Q - \lambda/d), & \gamma_Q \in (1, 3), \lambda/d \leq 1 + \gamma_Q \\ \emptyset, & \gamma_Q \in (1, 3), \text{else.} \end{cases} \quad (27)$$

We derive the quantization beam's direction from the conjugate equality with $d = \lambda/2$. Correspondingly, we have the following corollary.

Corollary 3: When $d = \lambda/2$, the quantization beam's direction ϕ_Q can be calculated by

$$\phi_Q = \begin{cases} \arccos(\gamma_Q + 2), & \gamma_Q \in (-3, -1) \\ \arccos(\gamma_Q - 2), & \gamma_Q \in (1, 3) \\ \arccos(\gamma_Q), & \gamma_Q \in (-1, 1) \\ 0 \text{ and } \pi, & \gamma_Q = -1, 1, -3 \text{ or } 3. \end{cases} \quad (28)$$

We need to emphasize that the generalization of Corollary 1-3 from ULA to URA is straightforward, which can be derived by leveraging the conjugate equality and similar derivation in Appendix A of [11].

IV. MULTI-DIRECTION BEAM SWITCHING

A. Problem Formulation

We propose to design the beam-switching codebook by maximizing the sum capacity from each direction of receivers. In particular, denote the codebook matrix as $\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_{N_{Rx}}]$, where $\mathbf{c}_i$ is the codeword for the i -th Rx and is generated from vector space of $\mathbf{w}_q$. Then, the MU capacity optimization problem can be formulated as

$$(P1): \max_{\mathbf{C}, \mathbf{z}} J(\mathbf{C}, \mathbf{z}) = \sum_{i=1}^{N_{Rx}} \log \left(1 + \frac{P_{Tx} \mathbf{c}_i^H \mathbf{R}_i \mathbf{c}_i}{\sigma_i^2} \right), \quad (29)$$

where

$$\mathbf{R}_i = \mathbf{a}_C(\Omega_{out,i}, \Omega_{in,Tx}) \mathbf{a}_C(\Omega_{out,i}, \Omega_{in,Tx})^H, \quad (30)$$

⁴Though generalization of $w_m = e^{j\theta}$ or $e^{j(\theta+\pi)}$ breaks the conjugate equality in Eq. (25), the phase rotation of θ does not change the modulus of inner product results of Eq. (25) and Eq. (9) is still satisfied.

is the i -th covariance matrix. Moreover, σ_i^2 is the power of noise at the i -th Rx. By solving (P1), we can obtain an optimized codebook $\mathbf{C}_{opt}$ and a prephasing map vector $\mathbf{z}_{opt}$.

B. Map Fusion Method

Since non-convexity is introduced by the integer variable, solving (P1) makes it hard to get an analytical optimality. We then propose a heuristic method to solve (P1) by adding certainty to the directions of candidate receivers while keeping the randomness for suppressing the quantization beam.

In particular, we obtain the codebook and prephasing map by solving N_{Rx} prephasing maps first for each Rx transmission. Then, N_{Rx} prephasing maps are merged into one final prephasing map by taking the mode (the single value that appears most often in a set of values) of N_{Rx} prephasing maps. Specifically, we denote the prephasing map matrix as

$$\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_{N_{Rx}}] \in \mathbb{C}^{M \times N_{Rx}} \quad (31)$$

where $\mathbf{z}_i, i = 1, 2, \dots, N_{Rx}$ is the prephasing map vector for the i -th Rx. Then, we obtain a fused map vector $\mathbf{z}^{opt}$ by taking the mode along each row vector in $\mathbf{Z}$, i.e.,

$$\mathbf{z}^{opt} = \text{mode}(\mathbf{Z}). \quad (32)$$

Note that the operation above can also suppress the power of the quantization beam since the mode-taking operation can bring randomness by itself.

To obtain the prephasing map vector $\mathbf{z}_i$ for the i -th Rx, we calculate quantized weights from MRC on a traditional 2-bit RIS. In particular, the optimal MRC weights vector $\mathbf{w}_{MRC, :, i}$ is calculated first, and then the quantized weights vector is obtained by taking the quantized phase closest to the continuous phase of MRC. We denote the operation to obtain $\mathbf{w}_{q, :, i}$ as

$$\mathbf{w}_{q, :, i} = Q(\mathbf{w}_{MRC, :, i}, [\mathbf{z}_{zeros}, \mathbf{z}_{ones}]), \quad (33)$$

where $\mathbf{z}_{zeros}$ and $\mathbf{z}_{ones}$ are all-zero and all-one vectors, respectively, and $\mathbf{z}_{zeros}/\mathbf{z}_{ones}$ represents a different phase space of 1-bit, as shown in Eq. (4). Thus, the concatenated term $[\mathbf{z}_{zeros}, \mathbf{z}_{ones}]$ means a feasible phase space of 2-bit RIS. Then, we obtain $\mathbf{z}_i$ by judging whether the phase value on the m -th element $\mathbf{w}_{q, m, i}$ belongs to type-1 or type-2. If $\mathbf{w}_{q, m, i} = \pm \frac{\pi}{2}$, then we assign 1 to $\mathbf{z}_{i, m}$, else we assign 0 to $\mathbf{z}_{i, m}$. Accordingly, we denote the operation to obtain $\mathbf{z}_i$ as

$$\mathbf{z}_i = \text{Map}(\mathbf{w}_{q, :, i}). \quad (34)$$

After $\mathbf{z}^{opt}$ is obtained via Eq. (32), the next step is to design codewords. As $\mathbf{z}^{opt}$ is fixed, the available value range of each codeword $\mathbf{c}_i$ is also determined. We obtain $\mathbf{c}_i$ by choosing its closest available values to $\mathbf{w}_{MRC, :, i}$, i.e.,

$$\mathbf{c}_i = Q(\mathbf{w}_{MRC, :, i}, \mathbf{z}^{opt}). \quad (35)$$

The computation of the proposed MF method contains a mode-taking operation, MRC weights computing and three quantization operations. Thus, the computation complexity can be denoted as $\mathcal{O}(N_{Rx}M)$.

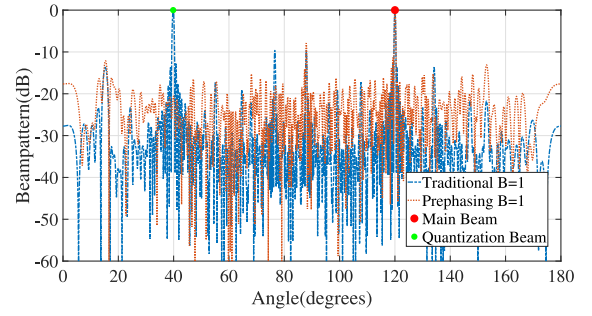


Fig. 2. The beam pattern of traditional 1-bit RIS and 1-bit prephased RIS, where $M = 256$.

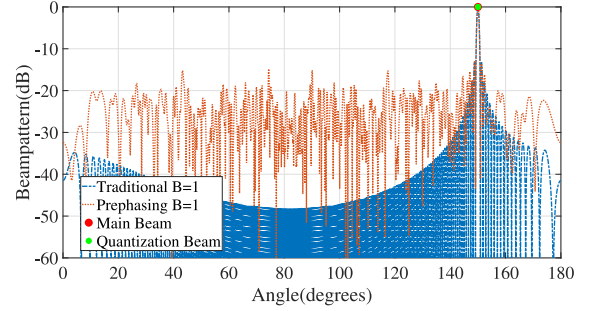


Fig. 3. The beam pattern of traditional 1-bit RIS and 1-bit prephased RIS, where $M = 256$ given specular reflection condition.

V. SIMULATIONS AND DISCUSSIONS

In simulations, the RIS is configured as a ULA with $M = 256$ elements and a half-wavelength spacing under 30 GHz frequency. The noise power is assumed to be -10 dBm for all receivers. Other specific parameters are introduced respectively for different simulations.

We first verify the proposed analysis to quantization beam and show the pros and cons of the prephasing method. In Fig. 2, we set a single transceiver pair location at $(\phi_{in}, T_x, \phi_{out,1}) = (30^\circ, 120^\circ)$. By Corollary 3, the quantization beam is analytically calculated at 39.829° , which is close to the simulated position of the quantization beam at 39.833° . In addition, the negative impact of applying random prephasing can be significant when specular reflection is considered on RIS. Namely, if we set $(\phi_{in}, T_x, \phi_{out,1}) = (30^\circ, 150^\circ)$, which is a special case⁵ as shown in Fig. 3, the random prephasing method only increases the sidelobe's power. In this case, both patterns show single-beam characteristics. Nevertheless, Corollary 3 holds in this scenario as two directions from the main beam and the quantization beam overlap with each other.

To further validate the accuracy of the derived quantization beam's direction, a numerical analysis is made on the error between the analytical calculation result and the numerical peak searched result under 10^{-3} accuracy in degrees, as shown in Fig. 4. The direction of the quantization beam is calculated by fixing Tx at 30° while considering Rx at $90^\circ, 105^\circ, 120^\circ, 135^\circ, 150^\circ$ and 165° respectively. The error between analytical and simulated results shows a decreasing trend with

⁵In this case, the optimal weights vector and prephasing map vector belong to traditional 1-bit RIS.

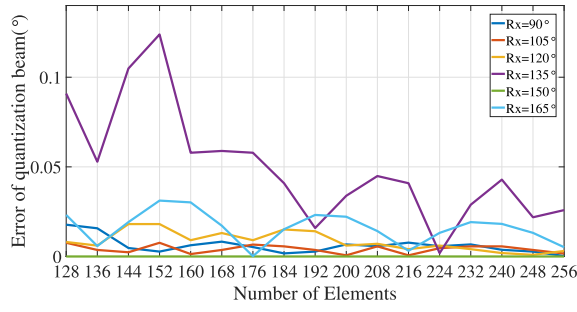


Fig. 4. The error of quantization beam between simulation and analytical result.

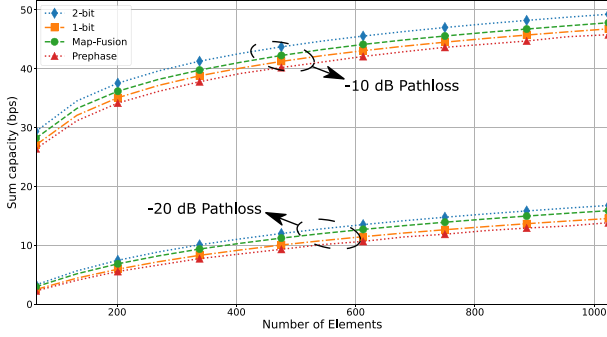


Fig. 5. Sum capacity given different path loss.

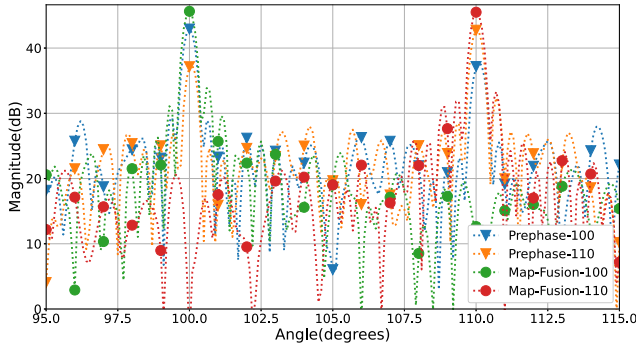


Fig. 6. A zoomed-in beam pattern of two code words given $N_{Rx} = 5$, shown from 95° to 115° .

the increase in the number of RIS elements. Though the error vary with incident direction and reflected direction, the maximal error indicated in Fig. 4 is less than 0.15° .

To evaluate the performance of MF method, the sum capacity is illustrated by RIS employing traditional 1-bit, 2-bit, prephasing and MF, as shown in Fig. 5. We set five directions of receivers that are linearly spaced between 100° and 140° while fixing Tx at 30° . The sum capacity is computed with the size of RIS, given -10dB and -20dB path loss in each LoS transmission. Fig. 5 shows that the sum capacity of using random prephasing is lower than that of using traditional 1-bit sum capacity. The reason is that the power gain is also severely sacrificed to suppress the quantization beam's power and keep the main beam's directivity. In addition, the MF outperforms traditional 1-bit on sum capacity, but the gap between the MF and 2-bit remains. In Fig. 6, the RIS

beam pattern of random prephasing is compared with that of MF, considering the same transceivers' direction set in Fig. 5. The beam pattern normalization to the amplitude is removed to see more specific power gains. Hence, quantization beams are mitigated by random prephasing and MF, and both methods result in single-beam patterns. Nevertheless, the proposed MF method outperforms the random prephasing method by around 2 dB in directions covered by codewords. This is because we leverage the degree of freedom, which exists within the randomness of prephasing, to enhance the power gain on passive beamforming of 1-bit RIS with two types of elements.

VI. CONCLUSION

In this letter, the quantization beam, an intrinsic issue in 1-bit RIS, and its direction in ULA are analyzed and mathematically derived. We formulate a beam-switching codebook design problem on 1-bit RIS with two types of phase-switching elements, which is solved by a heuristic MF method to maximize the sum capacity with the prephasing map vector and codewords. Simulation results verify the proposed corollaries of the quantization beam on 1-bit RIS with one type of elements and show that the MF method outperforms the random prephasing in the power gain of beam switching codebook while mitigating the quantization beam's power.

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